Readiness Inspection and Guidance Tool for Joule Scenarios

SAP S/4HANA Cloud Private Edition |FLP Plugin

Installation Guide

Step-by-step instructions for installing the tool on SAP S/4HANA Private Cloud Edition using abapGit.

1. Prerequisites

Before installing the Readiness Inspection and Guidance Tool for Joule Scenarios, ensure that the following prerequisites are met on your SAP landscape.

- Customer Managed Joule is deployed and set up on your landscape.
- SAP S/4HANA 2025 Private Cloud Edition Initial Shipment Stack i.e. ISS, FPS00 or higher.
- abapGit is installed on the SAP S/4HANA system. Refer to the documentation at Working with abapGit on the SAP Help Portal.
- The source code for the main report ZABAPGIT_STANDALONE is available in the abapGit git repository and can be downloaded from this link.

> **Remark:** The code must be deployed only on a sandbox or development system. This tool is intended for SAP S/4HANA Private Cloud Edition landscapes and should not be deployed on a Production system.

2. Setting Up the Root ABAP Package

To install the Readiness Inspection and Guidance Tool for Joule Scenarios, you must first create a root ABAP package that will serve as the container for all tool artefacts imported from the Git repository.

The package can be created using either ABAP Development Tools (ADT) in Eclipse or SAP GUI transaction SE21. Both methods achieve the same result. Choose your preferred method and follow the corresponding instructions below.

2.1 Method A: Using ABAP Development Tools (ADT)

- **Step 1:** Create a new Structure-type ABAP Package by opening the ***New ABAP Repository Object*** window (Ctrl + Shift + N).

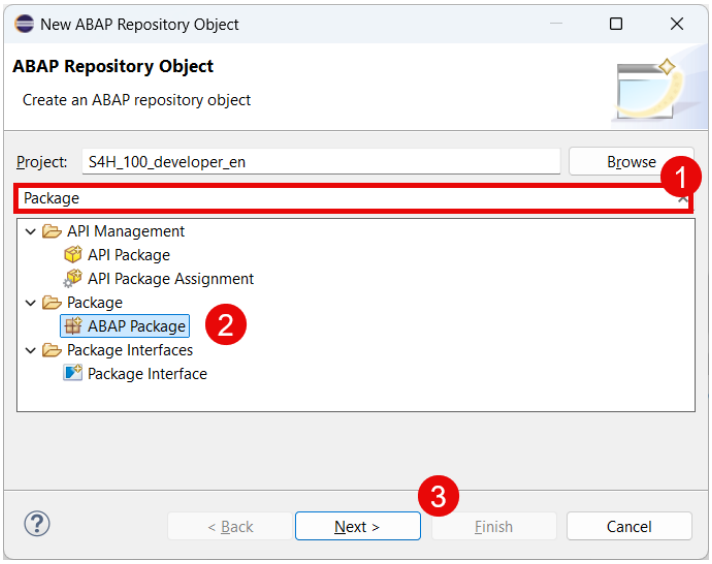


Figure: New ABAP Repository Object window (Ctrl+Shift+N)

- **Step 2:** Enter the package ***Name*** and ***Description*** in the New ABAP Package window:

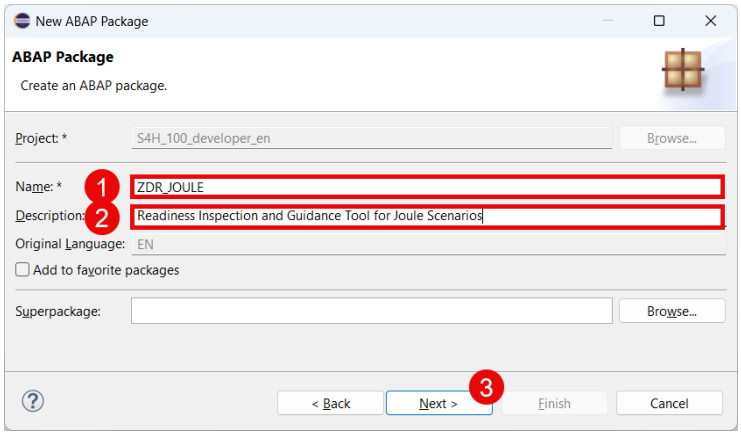


Figure: New ABAP Package window — enter Name and Description

- **Name =** ZDR_JOULE
- **Description =** Readiness Inspection and Guidance Tool for Joule Scenarios

- **Step 3:** Select **HOME** as the ***Software Component***, or choose another appropriate Software Component based on your deployment requirements, and click Next..

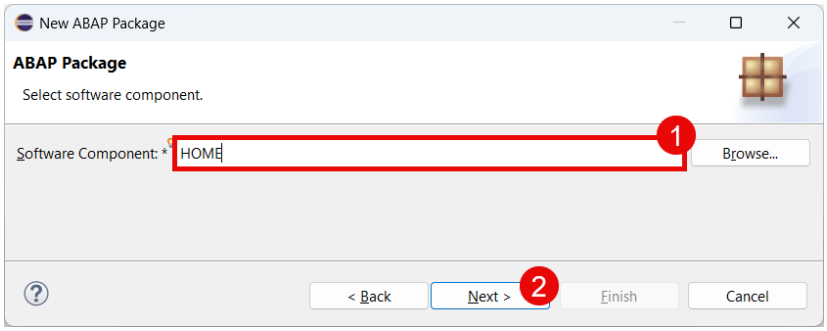


Figure: Select Software Component window — select HOME

- **Step 4:** On the Select Package Properties window, ensure that the ***Package Type*** is set to **Structure** and select the appropriate ***Transport Layer*** for your system.

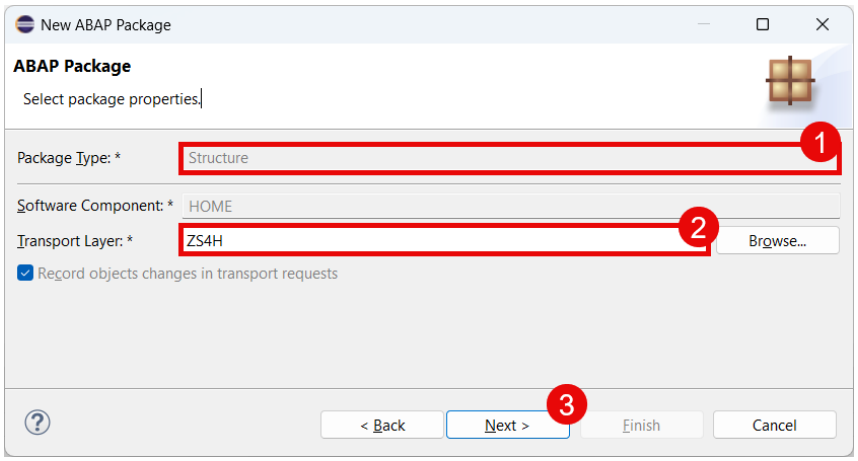


Figure: Select Package Properties — set Package Type to Structure

- **Step 5:** On the Select Transport Request window, choose Create a new request and enter a descriptive name, for example: Readiness Inspection Tool.

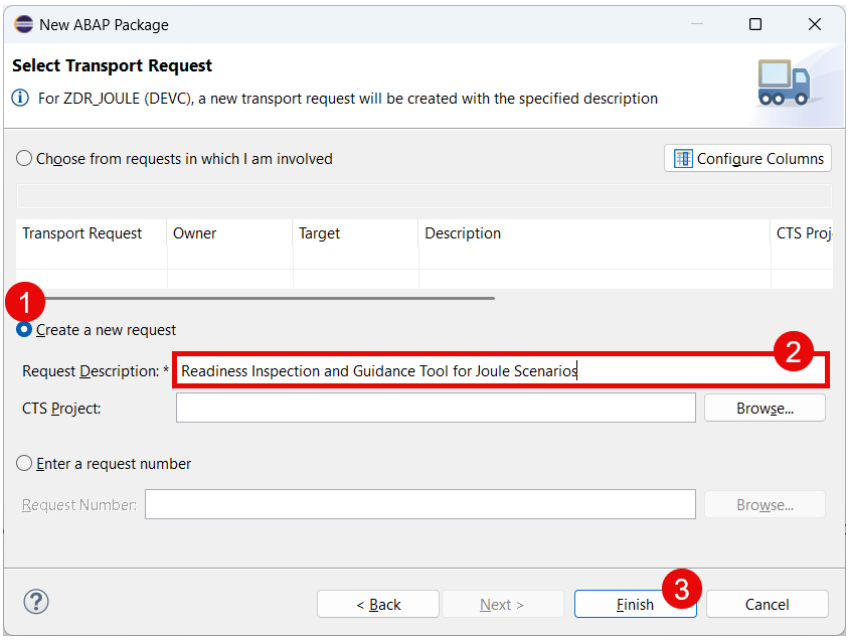


Figure: Select Transport Request — create a new request

> **Remark:** Double-check that the created package is of type Structure. This is critical for the correct deployment of the application artefacts.

> **Tip:** To locate the package and its artefacts more easily in the ABAP Development Tools, you can add the package to your Favorite Packages.

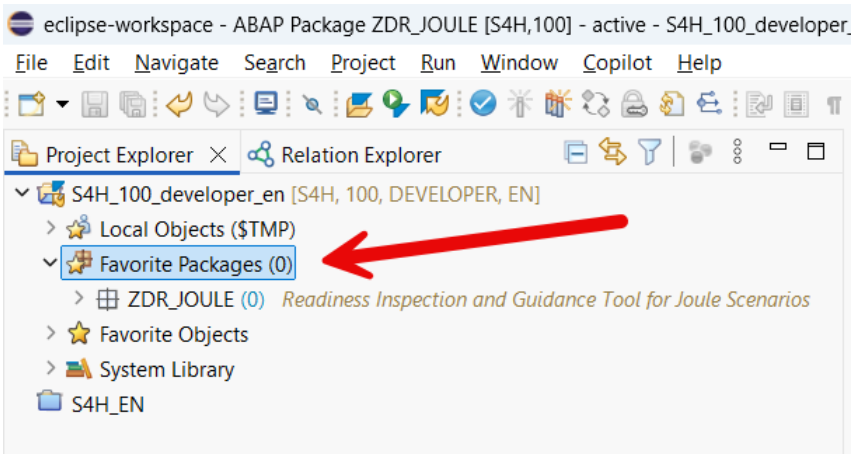


Figure: Package added to Favorite Packages for easy access

2.2 Method B: Using SAP GUI (SE21)

Step 1: Open Package Builder by executing the transaction **SE21**, select Package, enter **ZDR_JOULE** and click ***Create***.

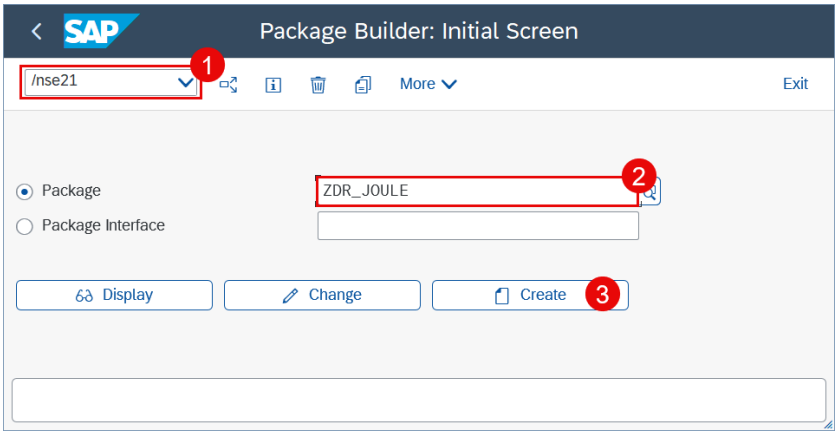


Figure: SE21 Create Package ZDR_JOULE

- **Step 2:** Enter Package Details and confirm:
  - **Name =** ZDR_JOULE
  - **Short Description =** Readiness Inspection and Guidance Tool for Joule Scenarios
  - **Software Component =** HOME (or choose an appropriate package based on your usage)
  - **Transport Layer =** Select the appropriate layer for your system
  - **Package Type =** Structure Package
  - **Superpackage =** Leave blank for the root package

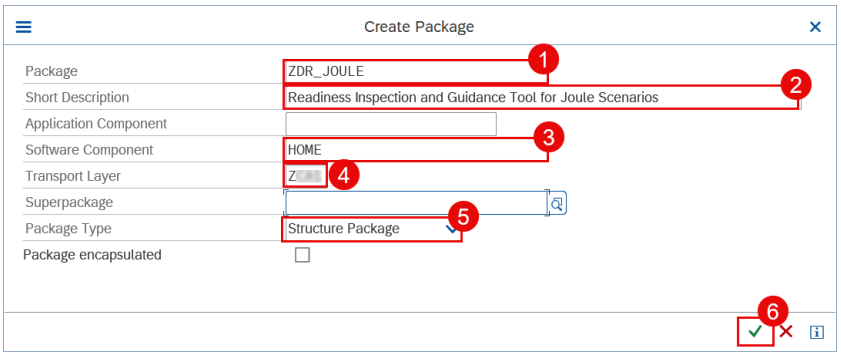


Figure: Enter Package Details

- **Step 3:** Assign an appropriate Workbench transport request, or create a new one if required, and click **Continue**.

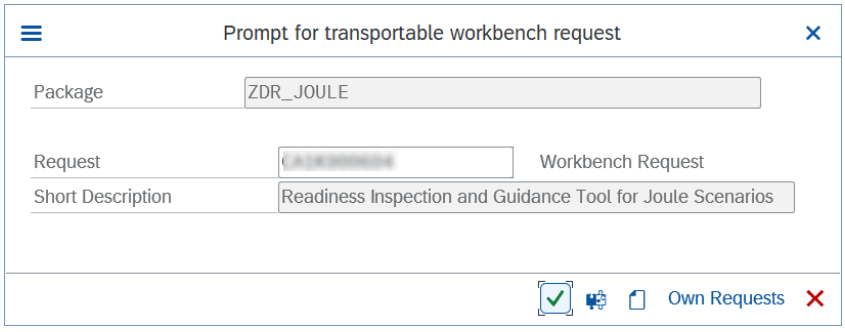


Figure: Transport Request Selection

3. Installing via abapGit

The Readiness Inspection and Guidance Tool for Joule Scenarios is hosted on GitHub as an open-source project. The following steps walk you through importing it into your SAP S/4HANA system using abapGit.

- **Step 1:** Ensure you can access the tool repository at: https://github.com/SAP/readiness-tool-for-joule
- **Step 2:** Open abapGit via transaction SE38 or SA38 by running the report ZABAPGIT_STANDALONE.

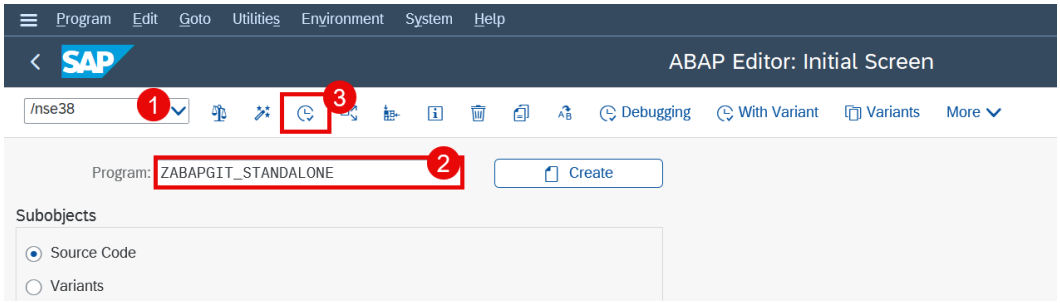


Figure: Running ZABAPGIT_STANDALONE via transaction SE38/SA38

- **Step 3:** Click the New Online button on the abapGit main screen.

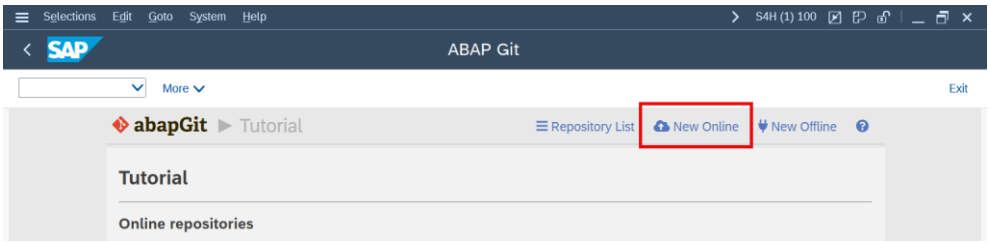


Figure: abapGit main screen — click New Online

- **Step 4:** Enter the following details on the New Online Repository screen:

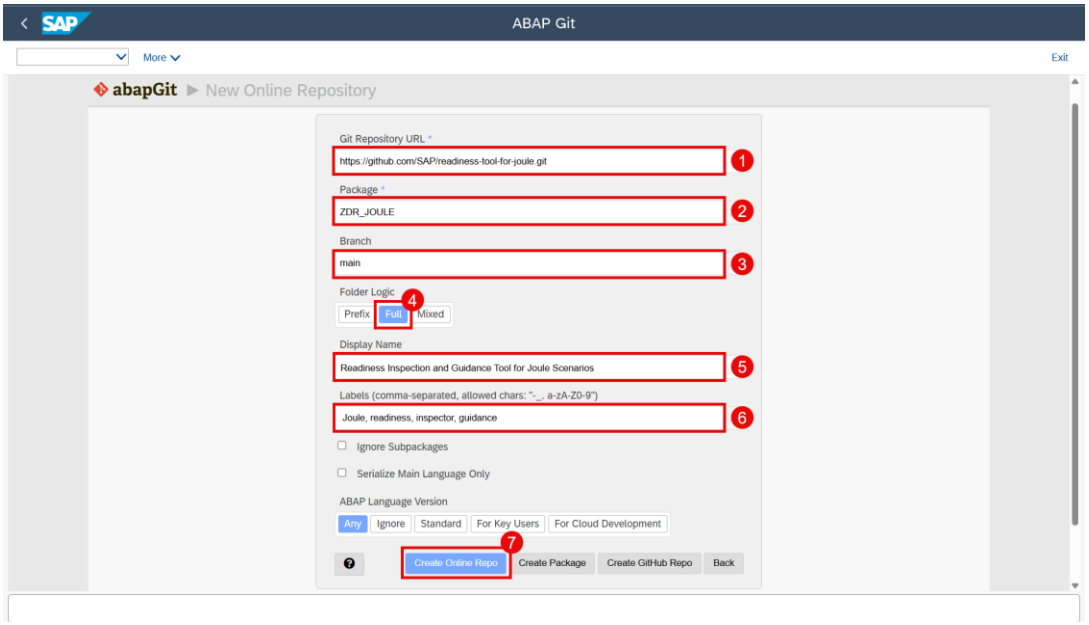


Figure: New Online Repository form — enter URL, package, and branch details

- **Git Repository URL =** https://github.com/SAP/readiness-tool-for-joule.git
- **Package =** ZDR_JOULE
- **Branch =** main
- **Folder Logic =** Full
- **Display Name =** Readiness Inspection and Guidance Tool for Joule Scenarios
- **Labels =** Joule, readiness, inspector, guidance?

- **Step 5:** Click Create Online Repo to register the repository in abapGit.
- **Step 6:** Log on to the Git repository using your credentials (GitHub username and personal access token or password) when prompted.

> **Note:** If you encounter a certificate error while connecting to the open-source repository, you will need to obtain the SSL certificate from your IT team and upload it to your SAP system via transaction STRUST.

- **Step 7:** abapGit will prepare the list of objects to import along with their current status. Review the list to confirm it is complete.

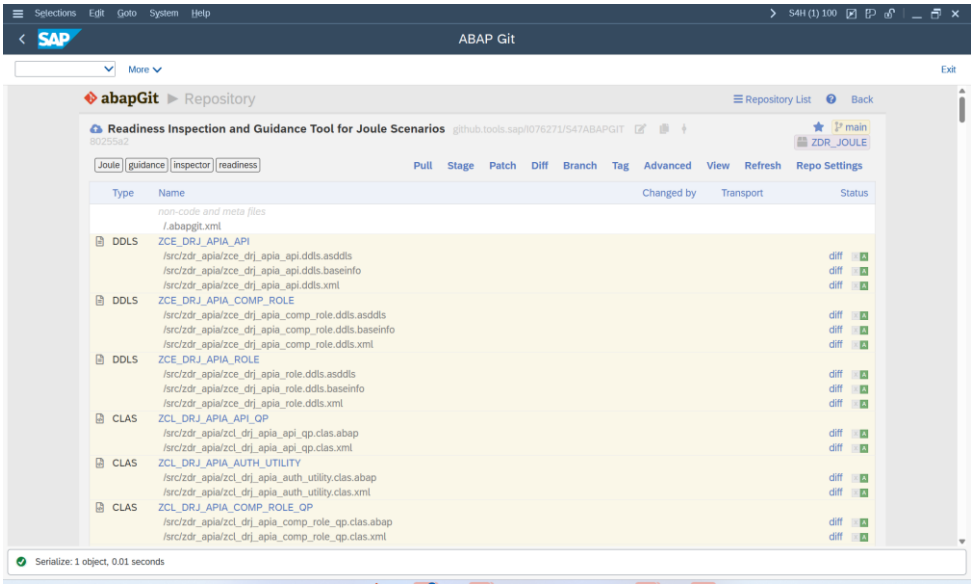


Figure: abapGit object list prepared for import

- **Step 8:** Click Pull to import the objects into your SAP S/4HANA system.

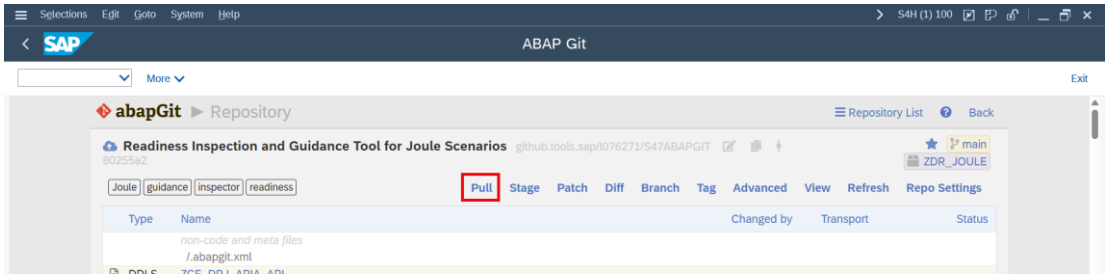


Figure: Click Pull to import objects into SAP S/4HANA

- **Step 9:** On the verification screen, abapGit displays all objects it will import. Review the list and click Continue to proceed.

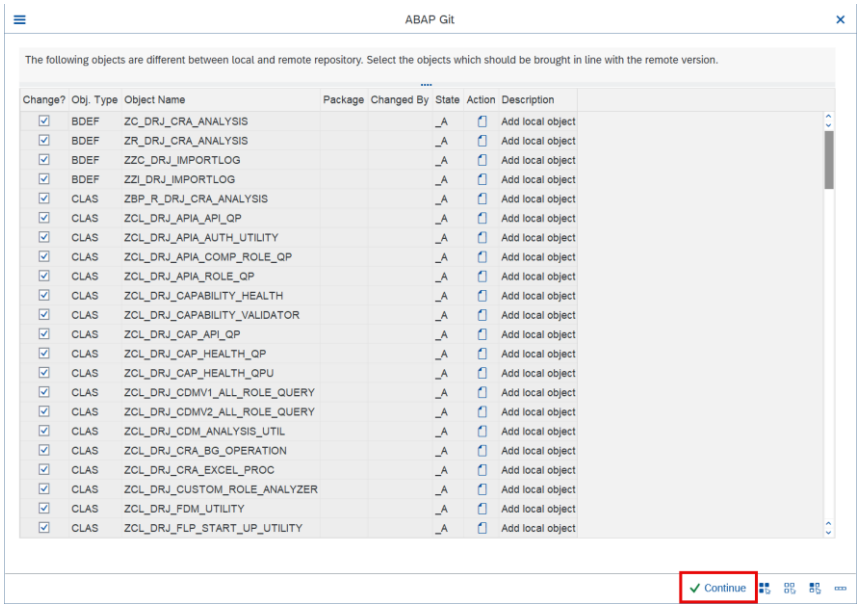


Figure: abapGit verification screen — State column should show _A for fresh install

> **Note:** For a fresh installation, the State column should show _A (Added) for all objects.

- **Step 10:** On the Prompt for Workbench Request screen, select the same transport request that was created during the ABAP package setup and confirm.

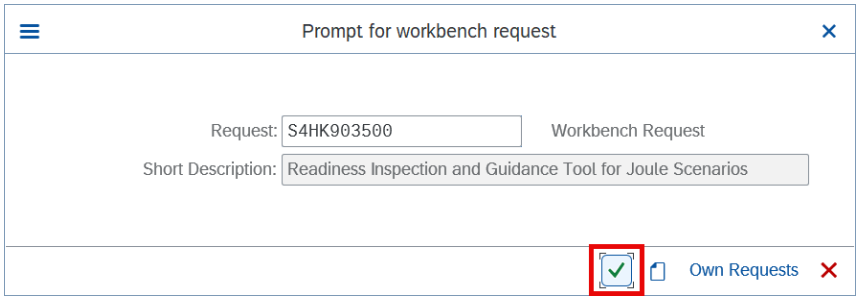


Figure: Prompt for Workbench Request — select the transport created earlier

- **Step 11:** abapGit will import all objects and attempt to activate them. On the Inactive Objects screen, locate the Service Binding entry **SRVB ZAPI_JOULECAPABILITY_O2**, uncheck it, and click **Continue**. The Service Binding must be activated manually in a separate step.

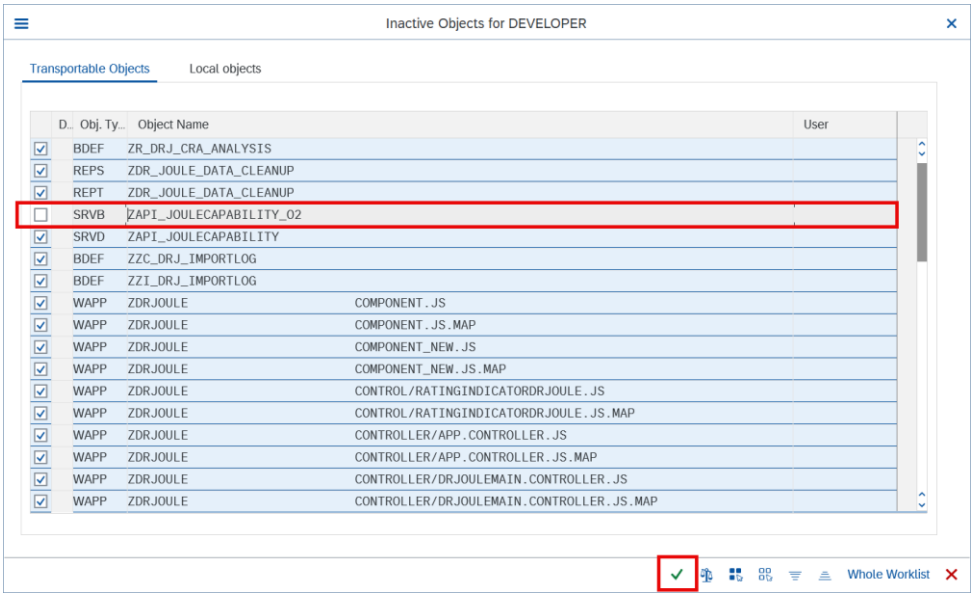


Figure: Inactive Objects screen — uncheck Service Binding SRVB ZAPI_JOULECAPABILITY_O2

- **Step 12:** After activation, you will see a single error related to the missing Service Binding. This is expected behavior — the Service Binding cannot be activated automatically by abapGit and will be resolved in the next section.

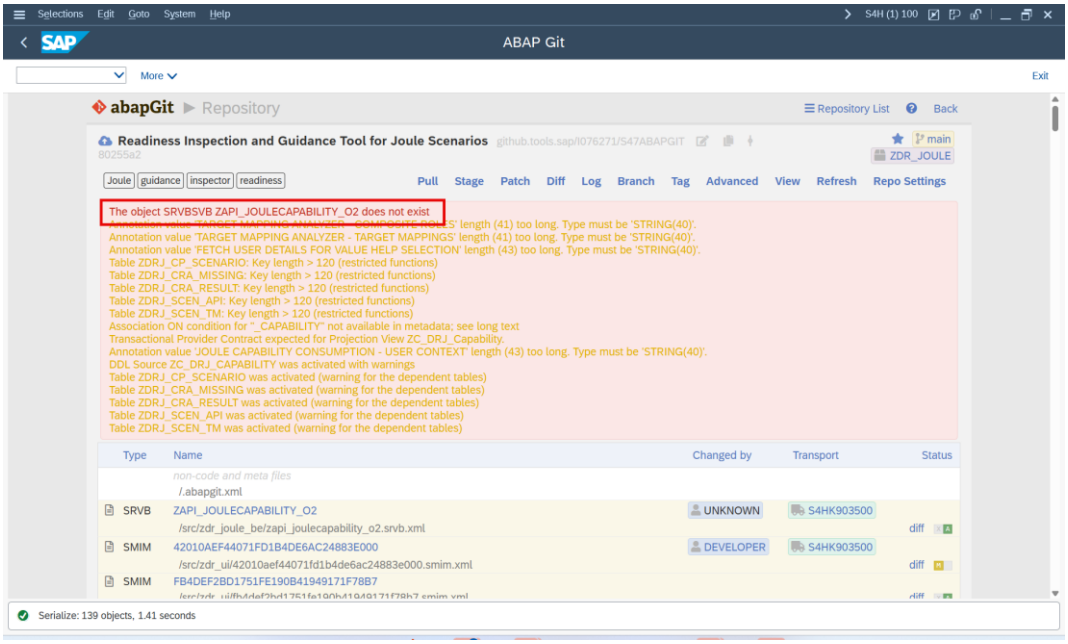


Figure: Expected single error — Service Binding must be activated manually

4. Activating the OData Service

The OData service that powers the tool requires two manual activation steps: activating the **Service Definition** and then activating the **Service Binding**. Both are performed in the ABAP Development Tools (ADT).

- **Step 1:** Open the ABAP Development Object window (Ctrl + Shift + A) and search for the Service Definition by name: ZAPI_JOULECAPABILITY.

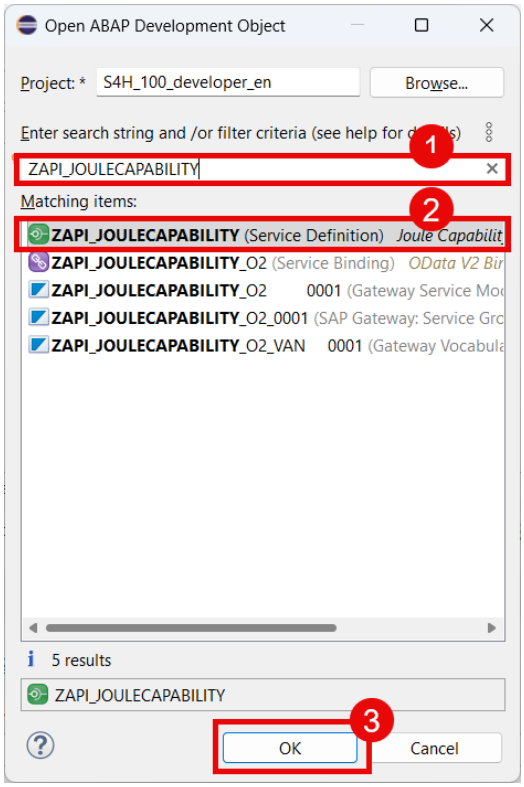


Figure: Open ABAP Development Object window — search for Service Definition

- **Step 2:** Click Activate to activate the Service Definition.

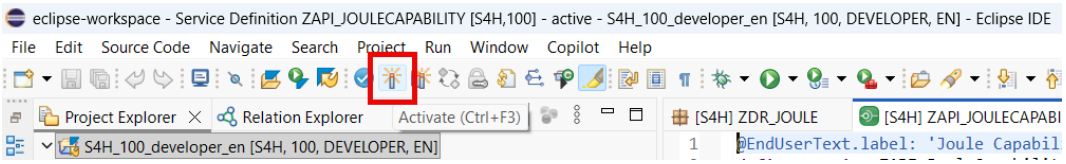


Figure: Activate the Service Definition ZAPI_JOULECAPABILITY_O2

- **Step 3:** Open the ABAP Development Object window again (Ctrl + Shift + A) and search for the Service Binding by name: ZAPI_JOULECAPABILITY_O2.

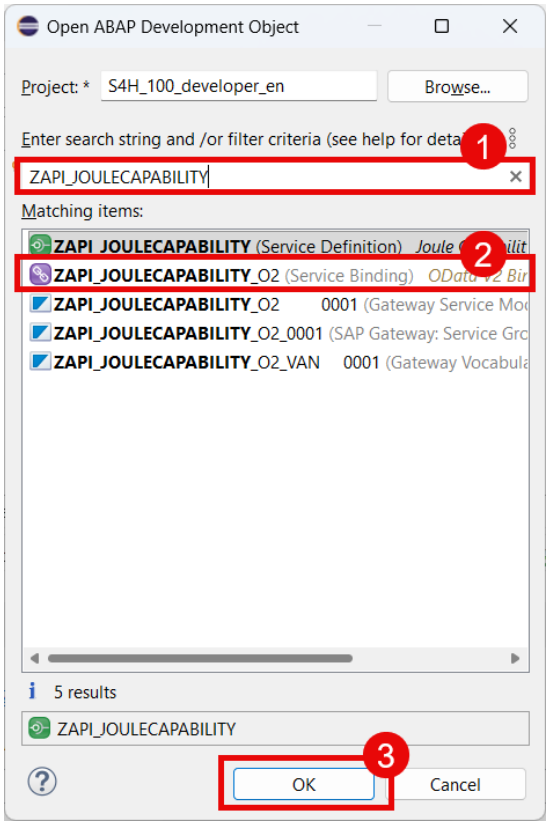


Figure: Open ABAP Development Object window — search for Service Binding

- **Step 4:** Click Activate to activate the Service Binding.

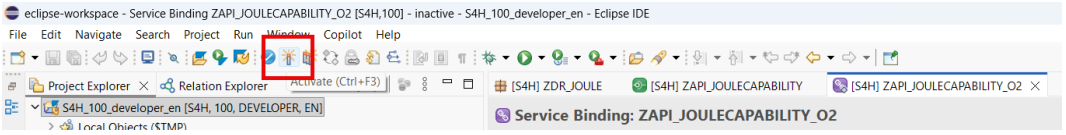


Figure: Activate the Service Binding ZAPI_JOULECAPABILITY_O2

- **Step 5:** Verify that the Service Binding is in Active state by checking the General properties tab.

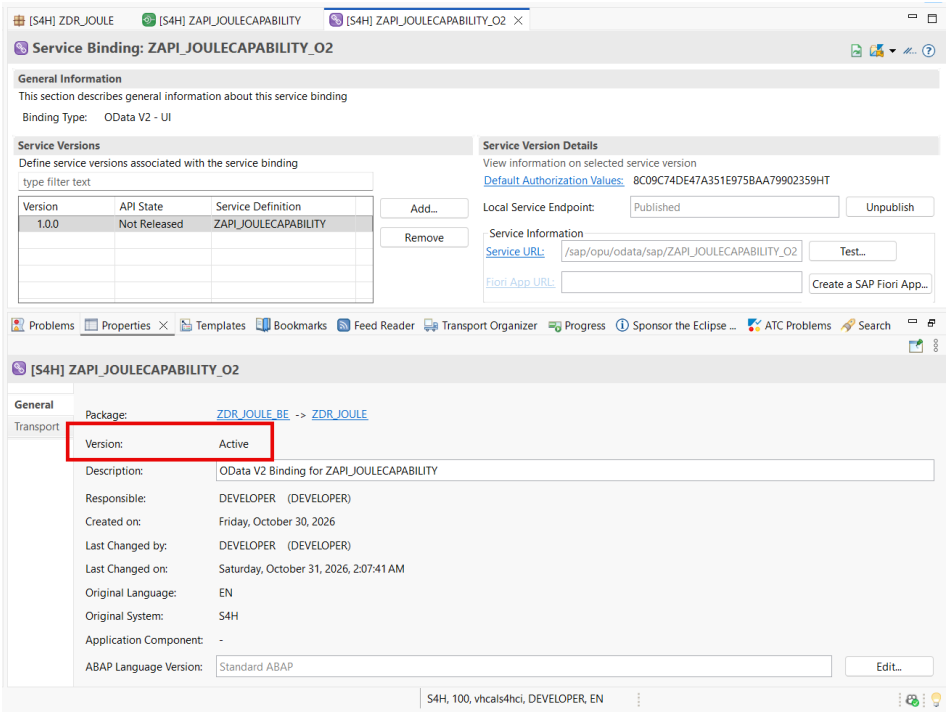


Figure: Service Binding General properties — verify Active state

- **Step 6:** Return to the abapGit screen and click the Refresh button to synchronise the object status.



Figure: abapGit screen — click Refresh after activating the Service Binding

- **Step 7:** After the refresh, the error "The object SREVBSVB ZAPI_JOULECAPABILITY_O2 does not exist" should no longer appear in the error log.

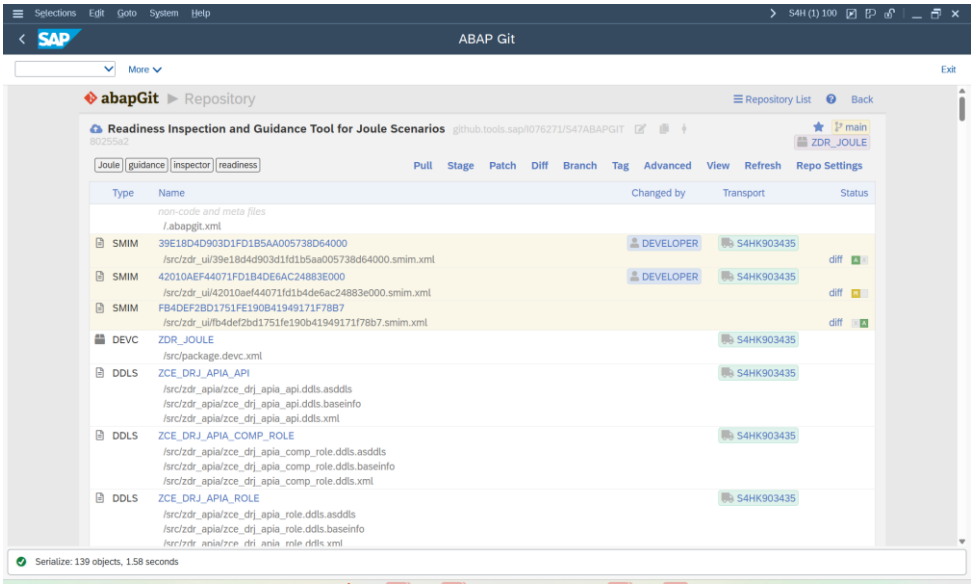


Figure: After refresh — Service Binding error has been resolved

> **Note:** Minor differences may appear for SMIM objects. These can be safely ignored and do not affect the functionality of the tool. The installation is considered complete at this point.

The installation also imports SAP Fiori Business Catalogs, which contain the Target Mappings required by the tool to launch from the SAP Fiori Launchpad.

5. Setting Up the PFCG Business Role

To grant users access to the tool in the SAP Fiori Launchpad, you need to upload and assign the PFCG Business Role that is shipped with the repository.

- **Step 1:** Download the PFCG role file Z_BR_JOULE_INSPECTOR.SAP from the root folder of the GIT repository to your local machine.
- **Step 2:** Open Role Maintenance via transaction PFCG. In the menu, click Role → Upload.

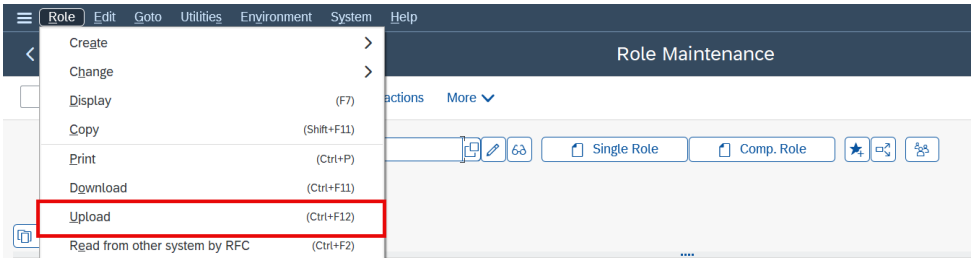


Figure: Role Maintenance (PFCG) — menu Role → Upload

- **Step 3:** Select the downloaded role file (Z_BR_JOULE_INSPECTOR.SAP) and click Continue on the List of Roles to Be Imported screen.

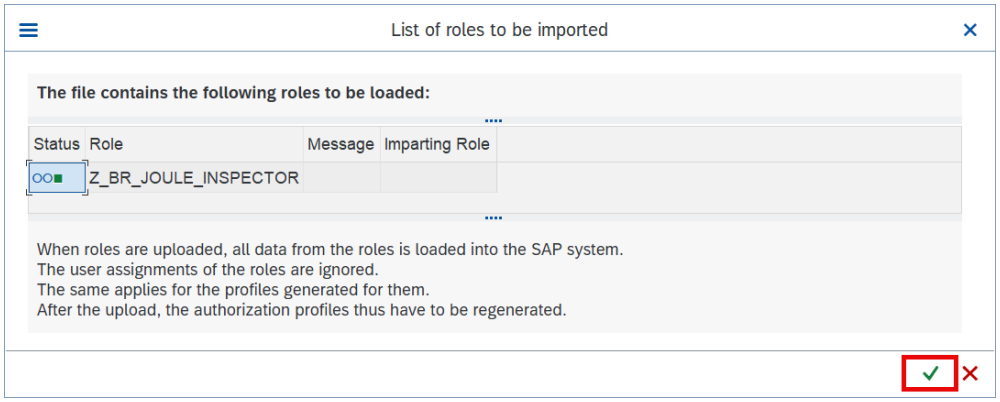


Figure: List of Roles to Be Imported — select the role and click Continue

- **Step 4:** After the role is imported successfully, click the Change button to open the role for editing.

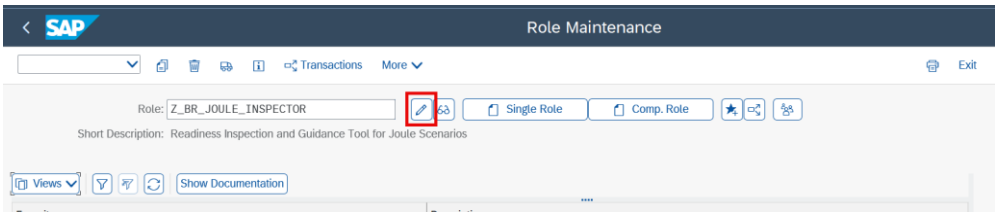


Figure: Imported role — click Change to open role maintenance

- **Step 5:** Click the Authorizations tab, then click Change Authorization Data under the Edit Authorization Data and Generate Profiles section. Click the Generate button to generate the authorization profile.
- **Step 6:** Go to the User tab, assign this role to the desired users, and click Save.

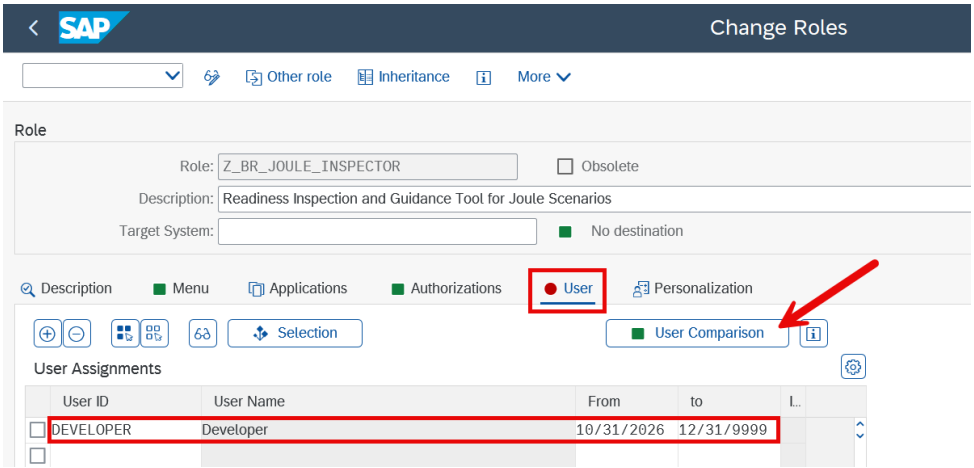


Figure: User tab — assign the role to users and click Save

- **Step 7:** If the User tab shows a red icon, click User Comparison to synchronise the user master records. After the comparison, the icon should turn green.

After completing the role assignment, users will see the Readiness Inspection and Guidance Tool for Joule Scenarios icon in their SAP Fiori Launchpad.

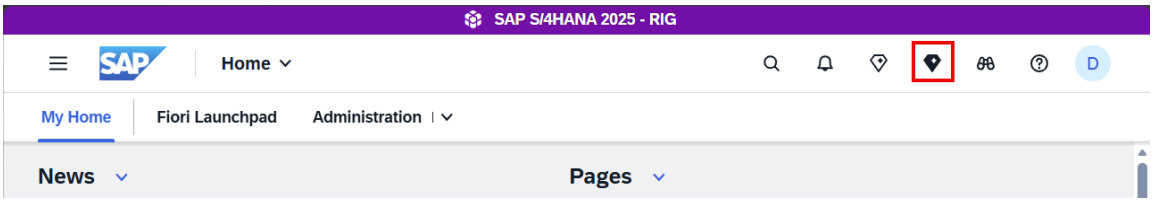


Figure: Readiness Inspection and Guidance Tool for Joule Scenarios icon in the SAP Fiori Launchpad